

**Optimizing AS-Net for MRI Brain Tumor Segmentation: A Comparative Study of
Different Encoders for Improved Segmentation Performance**

A Thesis by

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APPROVAL SHEET

This thesis entitled **Optimizing AS-Net for MRI Brain Tumor Segmentation: A Comparative Study of Different Encoders for Improved Segmentation Performance**, prepared and submitted by Mr. Shaun Niel Ochavo, and Mr. Armix Jay Roble in partial fulfillment of the requirements for the degree **Bachelor of Science in Computer Science** is hereby accepted.

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DEDICATION

To our families and friends, for showing us the true meaning of love and support. Your unwavering belief in us has been our greatest motivation. We dedicate this work to you, with all our love and gratitude.

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ABSTRACT

Accurate and efficient segmentation of brain tumors from Magnetic Resonance Imaging (MRI) is crucial for diagnosis and treatment planning. Deep learning models, particularly those incorporating attention mechanisms like the Attention Synergy Network (AS-Net), have shown promise. However, the original AS-Net, designed for skin lesion segmentation, utilized the computationally heavy VGG16 encoder. This study aimed to optimize AS-Net for MRI brain tumor segmentation by investigating the trade-off between segmentation accuracy and computational efficiency when replacing the VGG16 encoder with lightweight alternatives: MobileNetV3 (Large and Small) and EfficientNetV2 (Bo and B1). The core AS-Net architecture, featuring Spatial Attention (SAM), Channel Attention (CAM), and Synergy modules, was integrated with each encoder (all using a 224x224 input size except ENv2-B1 at 240x240) and evaluated on the BraTS 2023-GLI (Task 1) dataset, preprocessed into 2D HDF5 slices. Performance was assessed based on segmentation accuracy metrics (Dice Coefficient/F1 Score, IoU) and computational efficiency (total training time and average inference time per slice). Results evaluated on the test set demonstrated that the VGG16 baseline achieved the highest segmentation accuracy (Dice: 0.8873), very closely followed by EfficientNetV2-Bo (Dice: 0.8853) and EfficientNetV2-B1 (Dice: 0.8815). The lightweight models offered substantial efficiency gains, particularly in training time compared to VGG16 (e.g., MobileNetV3-Large trained approx. 5.4 hours faster [23% reduction], ENv2-Bo approx. 4.7 hours faster [20% reduction]), while inference times also showed improvements, albeit less pronounced. EfficientNetV2-Bo provided the best balance among the lightweight options, achieving accuracy nearly identical to VGG16 with significantly improved training efficiency and competitive inference speed. This research concludes that while VGG16 yielded the marginally highest accuracy in this setup, optimizing AS-Net with lightweight encoders like EfficientNetV2-Bo presents a highly viable trade-off for practical MRI brain tumor segmentation, balancing top-tier performance with enhanced computational efficiency.

Keywords: Brain Tumor Segmentation, MRI, Attention Synergy Network (AS-Net), Lightweight Encoders, Deep Learning, Medical Image Segmentation, Computational Efficiency, Training Time, Inference Time, Dice Coefficient, EfficientNetV2, MobileNetV3, VGG16, BraTS 2023.

CHAPTER 1. INTRODUCTION

1.1 Background of the Study

Medical image segmentation, particularly the delineation of brain tumors from Magnetic Resonance Imaging (MRI) scans, is a cornerstone of modern clinical oncology (S. Cai et al., 2022; X. Liu et al., 2021). Accurate segmentation provides critical information for diagnosis, treatment planning, and monitoring disease progression by precisely identifying tumor boundaries and characteristics (X. Liu et al., 2021; Tang, 2019). This process involves partitioning an image into distinct regions based on shared properties, thereby enhancing the interpretability of complex medical images and mitigating potential diagnostic errors (Handels et al., 2013; X. Liu et al., 2021).

The field of computer-aided diagnosis (CAD) has witnessed significant advancements, largely driven by the success of deep learning, especially Convolutional Neural Networks (CNNs). These techniques have demonstrated superior accuracy and robustness compared to traditional image processing methods for medical image segmentation tasks (Litjens et al., 2017). Architectures such as U-Net (Ronneberger et al., 2015) and its derivatives have become standard tools for segmenting organs and lesions across various imaging modalities, including MRI for brain tumor analysis (S. Cai et al., 2022; Moorthy & Gandhi, 2022; Tang, 2019). The automation offered by these deep learning models promises not only to improve the speed and consistency of tumor segmentation but also to alleviate the workload of radiologists and other healthcare professionals (Moorthy & Gandhi, 2022; Tang, 2019).

1.2 Statement of the Problem

Despite these strides, challenges remain. A recent advancement in image segmentation is the Attention Synergy Network (AS-Net), proposed by K. Hu et al. (2022) for

skin lesion segmentation. AS-Net effectively utilizes spatial and channel attention mechanisms to improve segmentation performance. However, the original study acknowledged significant limitations related to the computational demands and potential generalizability issues stemming from its reliance on the relatively heavy VGG16 encoder architecture. This computational inefficiency poses a barrier to practical deployment in clinical settings where resources may be constrained and rapid processing is often desirable.

Therefore, the central problem addressed by this research is the suboptimal computational efficiency of the original AS-Net architecture when considering its potential application to the demanding task of MRI brain tumor segmentation. While the attention mechanisms show promise, the VGG16 encoder limits its practicality. This study investigates whether adapting AS-Net with more modern, lightweight encoder architectures can maintain or improve segmentation accuracy while significantly enhancing computational efficiency.

This study seeks to answer the following research questions:

1. How can lightweight encoder architectures, specifically MobileNetV3 and EfficientNetV2, be effectively integrated into the existing AS-Net framework for MRI brain tumor segmentation?
2. How do the AS-Net variants using MobileNetV3 (Large and Small) and EfficientNetV2 (Bo and B1) encoders compare to the original AS-Net with a VGG16 encoder in terms of:
 - Segmentation accuracy (measured by Dice Coefficient/F1 Score, IoU/Jaccard Index, Precision, Recall, Binary Accuracy)?
 - Computational efficiency (measured by total training time)?
3. Does the synergy of spatial and channel attention mechanisms within AS-Net remain effective for segmenting brain tumors in MRI data when paired with different encoder backbones?

1.3 Objectives of the Study

The primary purpose of this research is to adapt and optimize the Attention Synergy Network (AS-Net) architecture for the task of MRI-based brain tumor segmentation by systematically comparing the performance and efficiency of different encoder backbones.

Specifically, this study aims to:

- Evaluate the integration of lightweight encoder architectures (MobileNetV3-Large, MobileNetV3-Small, EfficientNetV2-B0, EfficientNetV2-B1) into the AS-Net framework as alternatives to the original VGG16 encoder.
- Quantify and compare the segmentation performance (using metrics like Dice Coefficient, IoU, Precision, Recall) and computational efficiency (measured by training time) of the AS-Net variants equipped with these different encoders.
- Assess the effectiveness of the AS-Net's spatial and channel attention mechanisms when applied to the specific domain of MRI brain tumor segmentation.
- Provide evidence-based recommendations regarding the most suitable encoder architecture for balancing accuracy and efficiency within the AS-Net framework for this application.

1.4 Significance of the Study

This research holds significant potential for advancing knowledge in medical image analysis, particularly concerning deep learning applications for brain tumor CAD. Adapting and optimizing the AS-Net architecture (K. Hu et al., 2022) for MRI brain tumor segmentation contributes valuable insights into the design and application of attention-based deep learning models for complex medical imaging tasks.

The exploration of lightweight encoders like MobileNetV3 and EfficientNetV2 within the AS-Net framework directly addresses a critical need for practical, deployable AI solutions in healthcare. By demonstrating the feasibility of achieving high

segmentation accuracy with reduced computational cost, this study can facilitate the adoption of advanced segmentation tools in real-world clinical workflows, potentially operating on resource-constrained hardware or enabling faster processing times.

The findings are expected to benefit multiple stakeholders:

1. **Clinicians and Radiologists:** Access to more efficient and accurate automated segmentation tools can enhance diagnostic confidence, streamline treatment planning, and improve the monitoring of tumor response, ultimately supporting better patient management.
2. **Patients:** Improved accuracy and efficiency in segmentation can contribute to earlier detection, more precise interventions, and potentially improved prognoses.
3. **Researchers and Developers:** This work provides empirical evidence on the performance trade-offs of different encoders within an attention-based framework, informing future development of deep learning models for medical imaging. The adaptation of the attention synergy mechanism offers insights into the role of attention in extracting relevant features from MRI data.

Ultimately, this research aims to bridge the gap between cutting-edge deep learning techniques and practical clinical application, contributing to the development of more accessible and effective tools for brain tumor analysis.

1.5 Scope and Limitation of the Study

1.5.1 Scope

This research is specifically focused on:

- Adapting and evaluating the Attention Synergy Network (AS-Net) architecture.

- The task of binary brain tumor segmentation (tumor vs. background) using MRI data. Multi-class segmentation (differentiating tumor sub-regions) is outside the scope.
- Comparing the performance of five specific encoder backbones: VGG16, MobileNetV3-Large, MobileNetV3-Small, EfficientNetV2-B0, and EfficientNetV2-B1. Other potential encoders are not evaluated. Input sizes were 224x224 for VGG16, MobileNetV3-Large/Small, and EfficientNetV2-B0, and 240x240 for EfficientNetV2-B1.
- Utilizing the BraTS 2023-GLI (Task 1) dataset. The original NIfTI files were pre-processed into 2D HDF5 slices for efficient loading. The study focuses only on the glioma segmentation task (Task 1) from the BraTS 2023 challenge and does not involve acquiring raw MRI data.
- Evaluating performance based on standard segmentation metrics (Dice, IoU, etc.) and computational efficiency based on training time on specific hardware.

1.5.2 Limitations

The study acknowledges the following limitations:

- **Dataset Specificity:** Findings are based on the BraTS 2023-GLI (Task 1) dataset. The generalizability of the results to other datasets (including other BraTS 2023 tasks) with different patient populations, tumor types, or imaging protocols requires further investigation.
- **Data Characteristics:** The quality, resolution, and potential artifacts within the original NIfTI files, as well as the specific method used for preprocessing into HDF5 slices, could influence model performance.
- **Computational Resources:** Training and evaluation were conducted on specific hardware (NVIDIA Tesla V100). Training times are relative and would dif-

fer on other systems. The scale of experimentation (e.g., number of runs, extent of hyperparameter tuning) was constrained by available resources.

- **Hyperparameter Selection:** The study used a fixed set of primary hyperparameters (e.g., learning rate, loss weights) across models for comparability. Individual tuning for each encoder might yield slightly different performance rankings.
- **Implementation Details:** Specific choices regarding fine-tuning strategy (which layers to freeze/unfreeze) and skip connection points could impact results.

CHAPTER 2. REVIEW OF RELATED LITERATURE

This chapter reviews the existing literature relevant to MRI brain tumor segmentation using deep learning, providing the context for the current study. It begins by discussing the importance and challenges of brain tumor segmentation using MRI. It then covers the evolution from traditional methods to deep learning approaches, focusing on CNNs and the U-Net architecture. Subsequently, it delves into the concept and application of attention mechanisms within deep learning for medical image segmentation. The chapter then specifically examines the Attention Synergy Network (AS-Net), detailing its architecture and identifying its limitations related to computational efficiency. Finally, it explores the significance and characteristics of lightweight encoder architectures, such as MobileNetV3 and EfficientNetV2, positioning them as potential solutions to overcome AS-Net's efficiency bottleneck for practical application in brain tumor segmentation.

2.1 MRI Brain Tumor Segmentation: Importance and Challenges

Magnetic Resonance Imaging (MRI) is a primary modality for diagnosing and assessing brain tumors due to its excellent soft-tissue contrast and ability to provide multi-parametric information through different sequences (e.g., T1, T1-contrast enhanced (T1ce), T2, FLAIR) (Gordillo et al., 2013). Accurate segmentation of brain tumors from MRI volumes is clinically crucial for several reasons: it aids in diagnosis and grading, guides surgical planning and radiotherapy treatment design, enables quantitative monitoring of tumor response to therapy, and facilitates volumetric analysis for tracking disease progression (Bakas et al., 2021; W.-L. Cai & Hong, 2018; Tang, 2019). Manual segmentation, however, is a laborious, time-consuming task requiring significant expertise, and suffers from inter- and intra-rater variability (Bakas et al., 2021). This motivates the development of automated and semi-automated segmentation methods.

Despite significant research, automated brain tumor segmentation remains challenging due to several factors (Gordillo et al., 2013; Litjens et al., 2017):

- **Tumor Heterogeneity:** Brain tumors, especially gliomas (the focus of the BraTS challenge), are highly heterogeneous in shape, size, location, and appearance across patients and even within the same tumor (e.g., necrotic core, enhancing tumor, edema) (Bakas et al., 2021; Ranjbarzadeh et al., 2021).
- **Ill-defined Boundaries:** Tumor boundaries can be infiltrative and indistinct, making precise delineation difficult.
- **Class Imbalance:** Tumor regions typically occupy a small fraction of the total brain volume, leading to highly imbalanced classes which can challenge standard training procedures (Havaei et al., 2017; Isensee et al., 2018).
- **Data Scarcity and Annotation Cost:** Acquiring large-scale, multi-modal MRI datasets with accurate, voxel-level ground truth annotations is expensive and time-consuming, limiting the data available for training supervised deep learning models (Ullah et al., 2023).
- **Computational Demands:** Processing large 3D MRI volumes, especially with complex deep learning models, requires substantial computational resources (GPU memory, processing power) (Havaei et al., 2017).

2.2 Evolution of Segmentation Methods

2.2.1 Traditional Approaches

Early attempts at automated brain tumor segmentation relied on traditional image processing techniques. These included intensity-based methods like thresholding (Gordillo et al., 2013), region-based methods like region growing (Melouah & Layachi, 2018), edge-based methods using active contours or level sets (Caselles et al., 1997; Pierre et al., 2021), and methods based on statistical models or machine learning classifiers using hand-crafted features (Gordillo et al., 2013). While useful, these

methods often struggled with the heterogeneity of tumors, required significant parameter tuning, and were sensitive to noise and imaging artifacts (W.-L. Cai & Hong, 2018).

2.2.2 Deep Learning and Convolutional Neural Networks (CNNs)

The advent of deep learning, particularly CNNs, has dramatically advanced the state-of-the-art in medical image segmentation (Litjens et al., 2017; X. Liu et al., 2021; Moorthy & Gandhi, 2022). CNNs automatically learn hierarchical features directly from image data, eliminating the need for manual feature engineering and capturing complex spatial patterns relevant for segmentation.

The U-Net architecture (Ronneberger et al., 2015) has been particularly influential in biomedical image segmentation. Its symmetric encoder-decoder structure with skip connections allows it to effectively combine high-level semantic context (from the encoder) with low-level spatial details (via skip connections), enabling precise localization and segmentation. Numerous variations and extensions of U-Net have been proposed for brain tumor segmentation, often incorporating deeper encoders, residual connections, or specialized modules (S. Cai et al., 2022; Havaei et al., 2017; Isensee et al., 2018). Other architectures like SegNet (Badrinarayanan et al., 2017) (using pooling indices) and DeepLab variants (L.-C. Chen et al., 2018) (using atrous convolutions) have also been explored.

Frameworks like nnU-Net (Isensee et al., 2021) represent the current benchmark in many medical segmentation tasks, including BraTS. nnU-Net automatically adapts the U-Net architecture and training pipeline based on dataset characteristics, achieving robust, state-of-the-art performance without manual architecture design. This highlights the power of well-configured deep learning pipelines but also underscores their potential complexity and resource requirements.

2.3 Attention Mechanisms in Medical Image Segmentation

To further improve the representational power of deep learning models, attention mechanisms (Vaswani et al., 2017) have been incorporated. Attention allows a network to dynamically weight the importance of different parts of the input data or feature maps, enabling it to focus on the most relevant information for the task at hand.

2.3.1 Types and Applications

Key attention types used in computer vision include:

- **Spatial Attention:** Focuses on 'where' the important information is located spatially in the feature map (Jetley et al., 2018).
- **Channel Attention:** Focuses on 'what' feature channels are most informative, adaptively recalibrating channel responses. The Squeeze-and-Excitation (SE) block (J. Hu et al., 2018) is a prime example.
- **Self-Attention/Transformers:** Models relationships between different positions in the input, capturing long-range dependencies (Vaswani et al., 2017). Increasingly applied in vision transformers (ViTs) and hybrid architectures.

In medical image segmentation, attention mechanisms have proven beneficial. The Attention U-Net (Oktay et al., 2018) introduced attention gates in skip connections to suppress irrelevant background features from the encoder before fusing them with decoder features. This improves focus on target structures like organs or tumors. Sinha and Dolz (2021) proposed a multi-scale self-guided attention strategy for U-Net, combining spatial and channel attention for brain tumor segmentation, demonstrating improvements over standard U-Net. These works show that guiding the network's focus via attention can lead to more accurate and robust segmentation, especially for complex structures like brain tumors (Ranjbarzadeh et al., 2021).

2.4 Attention Synergy Network (AS-Net)

Building on the success of attention mechanisms, the Attention Synergy Network (AS-Net) was proposed by K. Hu et al. (2022). Initially developed and evaluated for skin lesion segmentation, AS-Net introduces a novel way to combine spatial and channel attention within a U-Net like architecture.

2.4.1 Architecture and Synergy Mechanism

AS-Net utilizes an encoder (originally VGG16) and a decoder with skip connections. The core innovation lies within the decoder modules, which operate in parallel at each stage:

- **Spatial Attention Module (SAM):** Aims to highlight spatially salient regions. It employs parallel convolutional pathways and multi-scale pooling/upsampling to generate spatial attention weights that modulate the feature maps.
- **Channel Attention Module (CAM):** Focuses on inter-channel relationships. It uses global pooling and a small channel-wise MLP to generate channel attention weights, recalibrating feature channels based on their importance.
- **Synergy Module:** This module adaptively fuses the outputs from the parallel SAM and CAM branches using learnable scalar parameters (α and β). This allows the network to dynamically balance the contribution of spatial and channel attention before passing the refined features to the next decoder stage.

This synergistic combination aims to leverage the complementary strengths of both spatial and channel attention mechanisms throughout the decoding process.

2.4.2 Original Performance and Stated Limitations

K. Hu et al. (2022) reported state-of-the-art results for AS-Net on skin lesion segmentation benchmarks, demonstrating the effectiveness of the attention synergy approach compared to models using single attention types or simple concatenation.

However, the authors explicitly identified a key limitation: the reliance on the VGG16 encoder (K. Hu et al., 2022). VGG16 is known for its large number of parameters and high computational cost (FLOPs) compared to more modern architectures (Canziani et al., 2017). This inefficiency limits the model's inference speed, increases memory requirements, and makes training computationally expensive, potentially hindering its deployment in practical scenarios, especially outside its original domain. The authors suggested exploring more lightweight encoders as a direction for future work to address this efficiency concern (K. Hu et al., 2022).

2.5 Lightweight Architectures for Efficient Deep Learning

The need for computationally efficient deep learning models is paramount in medical imaging to enable deployment on resource-constrained hardware, facilitate faster processing, and reduce training costs (Canziani et al., 2017; Hirschorn et al., 2014; Ullah et al., 2023). Lightweight architectures are designed specifically to achieve a favorable balance between accuracy and efficiency (e.g., measured by parameters, FLOPs, latency, or training time).

2.5.1 MobileNetV3

MobileNetV3 (Howard et al., 2019) is a family of highly efficient CNNs optimized for mobile devices. Key features include:

- Depthwise separable convolutions to reduce computation.
- Inverted residual blocks with linear bottlenecks.
- Squeeze-and-Excitation (SE) modules for channel attention.
- Optimized network structures found via Neural Architecture Search (NAS), including efficient activation functions (h-swish).

It offers 'Large' and 'Small' variants catering to different performance/efficiency needs. MobileNet architectures have been explored for medical tasks, including

brain tumor segmentation with U-Net backbones (Pavithra et al., 2023), demonstrating their potential for efficiency.

2.5.2 EfficientNetV2

EfficientNetV2 (Tan & Le, 2021) improves upon the original EfficientNet family by focusing on both training speed and parameter efficiency. Its key contributions include:

- Combination of MBConv (used in EfficientNetV1) and Fused-MBConv blocks (more efficient early in the network).
- Progressive training strategy (increasing image size and regularization during training).
- Improved compound scaling strategy.

EfficientNetV2 models generally train faster and are more parameter-efficient than EfficientNetV1 and other contemporary models while achieving high accuracy. EfficientNet backbones have shown promise in medical segmentation tasks when integrated into U-Net architectures (Lin & Lin, 2024).

2.6 Recent Advances in Attention Mechanisms for Brain Tumor Segmentation

Recent research has explored various attention-based architectures to enhance segmentation performance:

- **Hybrid Multihead Attentive U-Net-3D:** Butt and Jabbar (2024) introduced a 3D U-Net variant incorporating multihead attention mechanisms, achieving superior segmentation accuracy on the BraTS 2020 dataset compared to traditional models like SegNet and Dense121 U-Net (Dice score improvement of 4.7%).
- **Aggregation-and-Attention Network (AANet):** Proposed by X. Wang et al. (2021), AANet integrates enhanced down-sampling, multi-scale connections,

and dual-attention fusion modules to improve feature representation, achieving 89.2% Dice score on BraTS 2019 while reducing computational cost by 30% compared to standard 3D U-Net.

- **MAG-Net:** Gupta et al. (2021) developed a multi-task attention-guided encoder-decoder network that simultaneously performs segmentation and classification of brain tumors, demonstrating improved performance with 40% fewer parameters than traditional architectures.
- **Multi-Threshold Attention U-Net (MTAU):** Awasthi et al. (2021) presented MTAU which employs multiple thresholds and attention mechanisms to segment different tumor regions, achieving Dice coefficients of 0.91, 0.87, and 0.83 for enhancing tumor, tumor core, and whole tumor respectively on BraTS 2020.
- **Cross-Modality Attention Fusion Network:** Y. Liu et al. (2023) proposed a novel attention mechanism that dynamically fuses features from different MRI modalities (T1, T2, FLAIR), improving segmentation accuracy by 3.5% compared to conventional fusion methods.
- **Recurrent Attention U-Net:** Zhao et al. (2022) introduced a recurrent attention mechanism that iteratively refines segmentation predictions, demonstrating particular effectiveness for small tumor regions often missed by single-pass attention models.

2.7 Advancements in Lightweight Architectures for Efficient Brain Tumor Segmentation

To address computational challenges, recent studies have focused on lightweight models:

- **EfficientNetV2 with Attention Mechanisms:** Zhang et al. (2024) enhanced EfficientNetV2 with global and efficient channel attention modules, achieving

92.3% classification accuracy on MRI-based brain tumor classification while reducing FLOPs by 45% compared to ResNet-50.

- **MobileNetV2-SVM Hybrid Model:** Research by Li et al. (2024) combined MobileNetV2 with a Support Vector Machine classifier, resulting in a model suitable for real-time applications with limited resources, achieving 88.7% accuracy on BraTS 2018 with inference times under 50ms per slice.
- **EfficientNetV2-M with Inception-V3:** H. Chen et al. (2024) proposed a concatenated model combining EfficientNetV2-M and Inception-V3 architectures, demonstrating enhanced classification performance on imbalanced datasets through transfer learning and data augmentation techniques (F1-score improvement of 7.2%).
- **Quantized Lightweight U-Net:** Khan et al. (2023) developed a 8-bit quantized version of U-Net with depthwise separable convolutions, achieving comparable performance to full-precision models while reducing memory requirements by 75%.
- **Neural Architecture Search for Medical Segmentation:** Y. Wang et al. (2023) employed NAS techniques to automatically design efficient architectures for brain tumor segmentation, discovering models that outperformed manually designed counterparts while using 60% fewer parameters.
- **Knowledge Distillation for Lightweight Models:** Yang et al. (2023) demonstrated how knowledge distillation from large teacher models could significantly improve the performance of lightweight student models for brain tumor segmentation, bridging the accuracy gap by up to 15%.

2.8 Research Gap and Contribution

While attention mechanisms and lightweight architectures have individually shown promise in brain tumor segmentation, their combined application within frameworks

like AS-Net remains underexplored. Huang et al. (2023) recently highlighted this gap in their comprehensive review of efficient medical image segmentation methods, noting that most attention-based models still rely on computationally heavy backbones. Similarly, Patel et al. (2024) identified encoder inefficiency as a major bottleneck in deploying attention-based segmentation models in clinical settings.

This study aims to fill this gap by systematically evaluating lightweight encoder alternatives (MobileNetV3 and EfficientNetV2) within the AS-Net framework. Our work extends beyond previous encoder comparison studies like Singh and Gupta (2023) by specifically focusing on the interaction between encoder choice and attention mechanism effectiveness. Furthermore, we address the clinical applicability concerns raised by Nguyen et al. (2024) by providing concrete metrics on both accuracy and computational efficiency using the latest BraTS 2023 dataset.

2.9 Definition of Terms

For clarity and consistency, the following key terms are defined as used in this thesis:

- **Magnetic Resonance Imaging (MRI):** A medical imaging technique that uses magnetic fields and radio waves to create detailed images of organs and tissues. Common modalities for brain tumors include T1-weighted (T1), T1-weighted post-contrast (T1ce), T2-weighted (T2), and Fluid-Attenuated Inversion Recovery (FLAIR).
- **Image Segmentation:** The process of partitioning a digital image into multiple segments or regions, often to locate objects and boundaries. In this context, it refers to identifying and delineating brain tumor regions from surrounding healthy tissue in MRI scans.
- **Convolutional Neural Network (CNN):** A class of deep neural networks, highly effective in computer vision tasks, that uses convolutional layers to automatically learn spatial hierarchies of features from input images.

- **Encoder-Decoder Architecture:** A common structure for segmentation networks (e.g., U-Net) where an encoder path progressively downsamples the input to capture context and a decoder path progressively upsamples to generate a high-resolution segmentation map.
- **Attention Mechanism:** A technique in deep learning that allows a network to dynamically weight the importance of different parts of the input data (e.g., spatial locations or feature channels) when making predictions.
- **AS-Net (Attention Synergy Network):** The specific deep learning architecture proposed by K. Hu et al. (2022), characterized by its synergistic use of spatial and channel attention modules within its decoder.
- **Encoder Backbone:** The initial part of a CNN (often pre-trained on a large dataset like ImageNet) used for feature extraction in architectures like AS-Net. Examples include VGG16, MobileNetV3, EfficientNetV2.
- **Lightweight Model:** A deep learning model designed to have significantly fewer parameters and lower computational requirements (FLOPs) compared to standard models, aiming for efficiency often suitable for mobile or edge devices.
- **VGG16:** A specific deep CNN architecture known for its simplicity and depth, but also its relatively high computational cost.
- **MobileNetV3:** A family of efficient CNN architectures designed for mobile devices, utilizing techniques like inverted residuals and squeeze-and-excitation blocks.
- **EfficientNetV2:** A family of CNN architectures achieving high accuracy and efficiency through compound scaling and optimized building blocks.
- **Dice Coefficient (F1 Score):** A metric used to evaluate the overlap between a predicted segmentation and a ground truth segmentation, ranging from 0

(no overlap) to 1 (perfect overlap). It is the harmonic mean of precision and recall.

- **IoU (Intersection over Union / Jaccard Index):** Another overlap metric for segmentation evaluation, calculated as the ratio of the intersection area to the union area of the prediction and ground truth.
- **BraTS Dataset:** The Brain Tumor Segmentation challenge dataset, a standard benchmark collection of annotated multimodal MRI scans used to evaluate and compare brain tumor segmentation algorithms in the research community.
- **FLOPs (Floating Point Operations):** A common metric for measuring computational complexity of deep learning models, representing the number of floating-point operations required to execute a forward pass through the network.
- **Inference Time:** The time taken by a trained model to make a prediction on new, unseen data, often measured per slice or per volume in medical imaging applications.
- **Precision:** A performance metric that quantifies the proportion of positive identifications that were actually correct, calculated as true positives divided by the sum of true positives and false positives.
- **Recall:** A performance metric that quantifies the proportion of actual positives that were correctly identified, calculated as true positives divided by the sum of true positives and false negatives.
- **Binary Accuracy:** The proportion of pixels correctly classified as either tumor or background in a segmentation task.
- **Fine-tuning:** The process of taking a model pre-trained on a large dataset (e.g., ImageNet) and further training it on a smaller, specific dataset (e.g., brain MRIs) for a related task, allowing knowledge transfer between domains.

- **Skip Connections:** Connections that bypass one or more layers in a neural network, typically used in encoder-decoder architectures to directly connect corresponding layers in the encoder and decoder paths, helping to retain fine-grained spatial information and mitigate the vanishing gradient problem.

CHAPTER 3. METHODOLOGY

3.1 Conceptual Framework

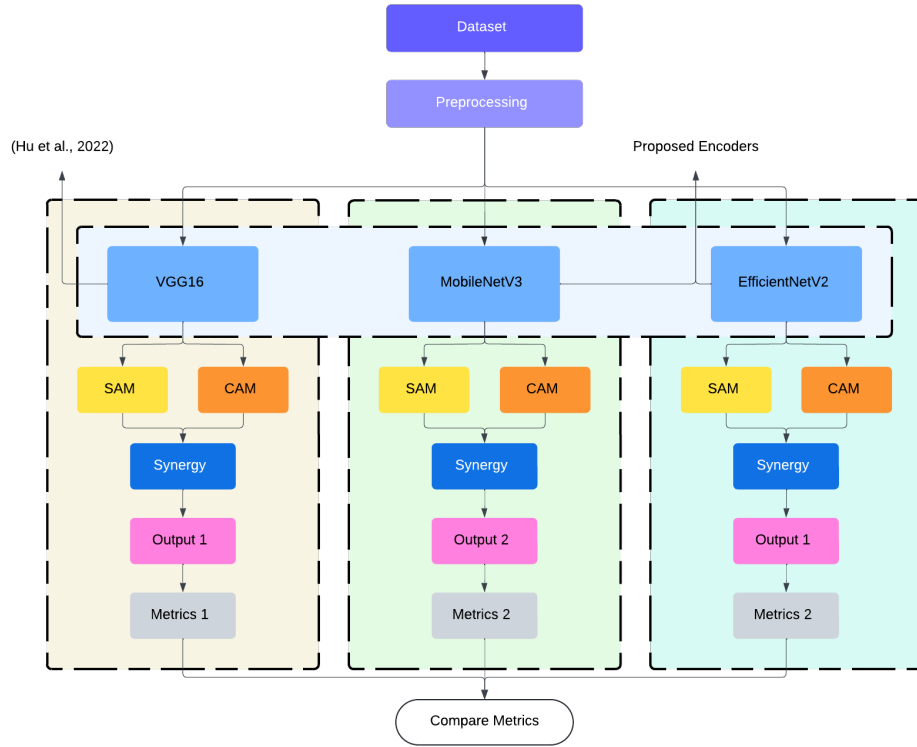


Figure 3.1: Conceptual framework

Figure 3.1 illustrates the conceptual framework guiding this research. The process initiates with the BraTS 2023-GLI (Task 1) dataset, which undergoes a standardized preprocessing pipeline. The resulting data is then fed into multiple parallel processing streams. Each stream implements the Attention Synergy Network (AS-Net) architecture but utilizes a distinct encoder backbone, such as VGG16, MobileNetV3, and EfficientNetV2, as explored in this study. Within each stream, the encoder's features are processed by Spatial Attention Modules (SAM) and Channel Attention Modules (CAM), whose outputs are integrated by a Synergy module. Each AS-Net variant generates its own segmentation output, from which performance metrics are calculated. Finally, these metrics are systematically compared across the differ-

ent encoder variants to evaluate their relative performance and efficiency.

3.2 The Attention Synergy Network (AS-Net) Architecture

The study adapts the AS-Net architecture, originally proposed by K. Hu et al. (2022) for skin lesion segmentation. The core components include:

- **Encoder Backbone:** Extracts hierarchical features from the input image. The original AS-Net used VGG16. This study replaces it with various alternatives.
- **Spatial Attention Module (SAM):** As illustrated in Figure 3.2, SAM processes input features through parallel convolutional pathways to generate spatial attention maps, highlighting salient regions (likely tumor areas) and focusing computation spatially.
- **Channel Attention Module (CAM):** As shown in Figure 3.3, CAM focuses on inter-channel relationships using global pooling and an MLP to generate channel-wise attention weights, emphasizing informative feature channels.
- **Synergy Module:** Integrates the outputs from SAM and CAM using learnable scalar parameters (α and β , initialized to 0.5), allowing adaptive fusion of spatial and channel attention before refinement and passing to the next decoder stage.
- **Decoder Structure:** Follows a standard U-Net-like symmetric structure, progressively upsampling features and integrating high-resolution spatial information from the encoder via skip connections at each stage. The fused features from the Synergy module are passed through the decoder stages.

3.3 Encoder Integration and Model Variants

Five pre-trained encoder networks were integrated into the AS-Net framework:

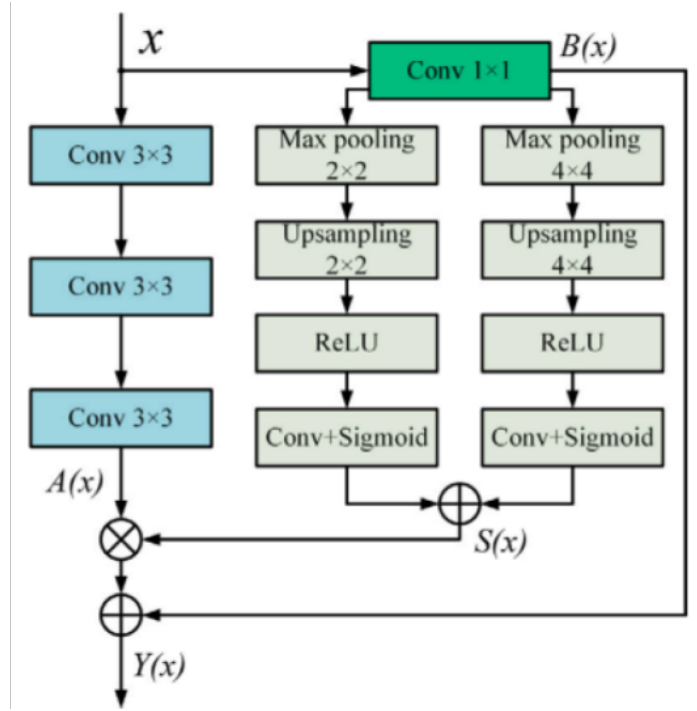


Figure 3.2: Diagram of the Spatial Attention Module (SAM) used in the AS-Net decoder. Adapted from K. Hu et al. (2022).

1. **Baseline Model (VGG16):** Used as the reference encoder, adapted from the original AS-Net paper (K. Hu et al., 2022). Input size: $224 \times 224 \times 3$.
2. **Lightweight Encoders:**
 - **MobileNetV3-Large & MobileNetV3-Small (Howard et al., 2019):** Selected for their high efficiency, designed for mobile applications, representing different points on the accuracy/latency curve. Input size: $224 \times 224 \times 3$.
 - **EfficientNetV2-Bo & EfficientNetV2-B1 (Tan & Le, 2021):** Chosen for their state-of-the-art balance of parameter efficiency and training speed, representing modern efficient architectures. Input sizes: $224 \times 224 \times 3$ (Bo), $240 \times 240 \times 3$ (B1).

These lightweight encoders were selected to address the computational limitations of VGG16 and evaluate modern efficient alternatives.

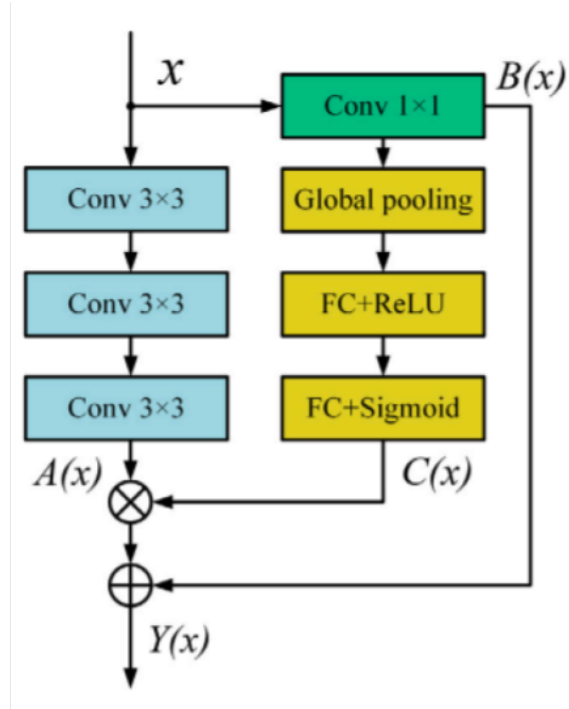


Figure 3.3: Diagram of the Channel Attention Module (CAM) used in the AS-Net decoder. Adapted from K. Hu et al. (2022).

3.3.1 Integration Process

Each encoder replaced the VGG16 feature extraction block. Skip connections were established by extracting feature maps from intermediate layers of each encoder that corresponded spatially to the resolution required at each stage of the AS-Net decoder. Specific layer choices were based on common practices for U-Net style architectures using these backbones (e.g., outputs of blocks ending with stride 2). Since encoders produce features with varying channel depths, 1x1 convolutional layers were potentially used within the decoder's skip connection pathway (before concatenation) if necessary to standardize channel dimensions, although the primary adaptation focused on selecting appropriate layers. For MobileNetV3 and EfficientNetV2, their respective built-in `preprocess_input` functions were applied within the model. For VGG16, Z-score normalization served this purpose. A custom `ResizeToMatchLayer` was used in the EfficientNetV2 decoders to ensure spatial dimension compatibility before concatenation, addressing potential minor discrepancies from padding/striding.

3.3.2 Transfer Learning Strategy

All encoders were initialized with weights pre-trained on ImageNet. A fine-tuning approach was employed:

- Initial convolutional blocks of each encoder were frozen (weights not updated).
- Later blocks were unfrozen and trained. The transition point varied per architecture (e.g., starting from 'block4' for VGG16, 'expanded_conv_6' for MobileNetV3-Large, 'block4' for EfficientNetV2 variants).
- The entire network (unfrozen encoder parts and the decoder) was trained end-to-end from the start.

This strategy aimed to leverage general features while adapting deeper layers to the MRI segmentation task.

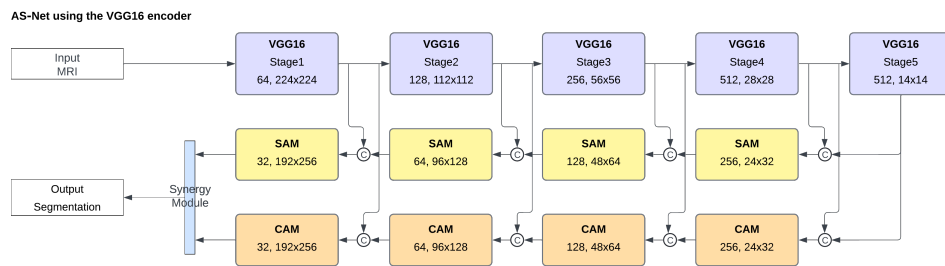


Figure 3.4: High-level architecture of AS-Net using the VGG16 encoder backbone.

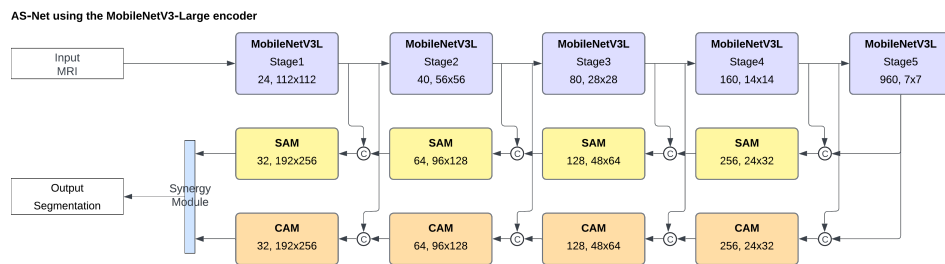


Figure 3.5: High-level architecture of AS-Net using MobileNetV3 (Large/Small) encoder backbones. Example shown uses MobileNetV3-Large.

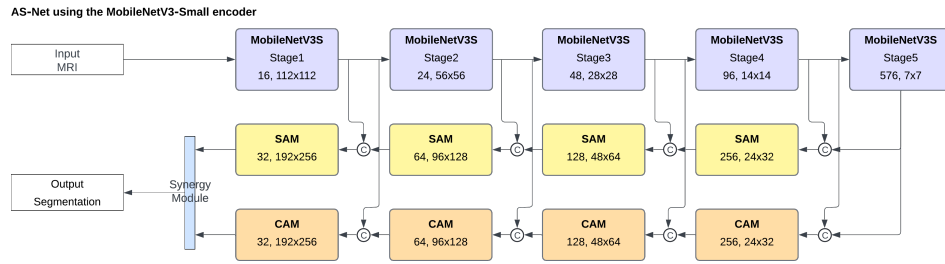


Figure 3.6: High-level architecture of AS-Net using the MobileNetV3-Small encoder backbone.

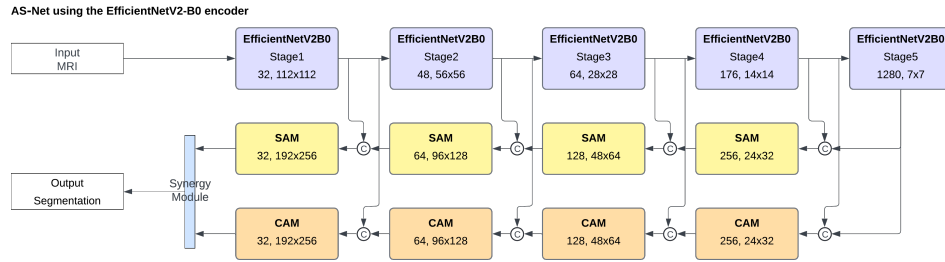


Figure 3.7: High-level architecture of AS-Net using the EfficientNetV2-B0 encoder backbone.

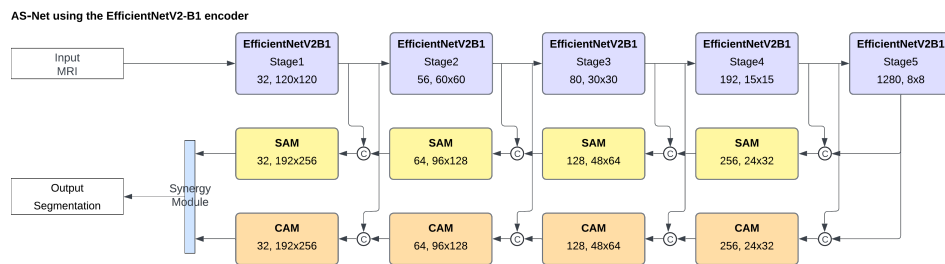


Figure 3.8: High-level architecture of AS-Net using the EfficientNetV2-B1 encoder backbone.

3.4 Dataset and Preprocessing

3.4.1 Dataset Identification

The study utilized the BraTS 2023-GLI (Task 1) (Brain Tumor Segmentation) challenge training dataset (Baid et al., 2021; Bakas et al., 2017a; Menze et al., 2014). Task 1 focuses on glioma segmentation. The dataset is hosted on the Synapse platform.

3.4.2 Ethical Considerations and Data Access

Access to the BraTS 2023 dataset requires registration and adherence to specific usage agreements, ensuring responsible use of sensitive medical data. The dataset

consists of anonymized MRI scans, protecting patient privacy. The steps followed to gain access were:

1. Registration for the BraTS 2023 challenge via the Synapse platform.
2. Completion and submission of the official BraTS Data Access form, providing a Synapse username and associated email.
3. Acceptance of an email invitation to join the BraTS 2023 Data Access Team on Synapse.

This process ensures that researchers agree to the data usage terms before downloading the dataset from Synapse. All data handling in this study complied with these terms. Researchers using this dataset are required to cite the relevant BraTS publications (Baid et al., 2021; Bakas et al., 2017a, 2017b, 2017c; Menze et al., 2014).

3.4.3 Original Data Description

The original dataset provides multi-modal 3D MRI scans (T1, T1ce, T2, FLAIR) in NIfTI format, along with expert-annotated ground truth segmentation masks for tumor sub-regions (NCR/NET, ED, ET). This study focused on creating a binary segmentation target (tumor vs. background).

3.4.4 Preprocessing Pipeline

A custom script converted the 3D NIfTI volumes into a collection of 2D HDF5 (H5) files, each representing a single axial slice. This facilitated efficient 2D processing. The pipeline included:

1. **File Path Handling & Metadata:** Read file paths from a metadata CSV mapping patient IDs, slice indices, and H5 paths.
2. **H5 File Structure:** Each H5 file contained:
 - **image:** Numpy array (e.g., 240x240x4) with T1, T1ce, T2, FLAIR data for the slice.

- `mask`: Numpy array (e.g., 240x240x3) with ground truth masks (NCR/NET, ED, ET).
3. **Data Splitting:** The list of H5 files was randomly shuffled (fixed seed) and split into training (80%), validation (10%), and test (10%) sets based on slices, not patients.
 4. **H5 Parsing:** Loaded `image` and `mask` arrays using TensorFlow's `tf.data` API.
 5. **Modality Selection:** Selected T1ce (index 1) and FLAIR (index 3).
 6. **RGB Channel Mapping:** Mapped [T1ce, FLAIR] to 3 channels using indices [0, 1, 0] -> [T1ce, FLAIR, T1ce].
 7. **Normalization:** Applied Z-score normalization independently to T1ce and FLAIR channels *before* RGB mapping (per-slice, per-channel mean/std).
 8. **Binary Mask Creation:** Converted the 3-channel ground truth `mask` to a single-channel binary mask (1 if any tumor label present, 0 otherwise). This was the training target.
 9. **Multi-Class Mask Preparation (for visualization):** Created a multi-class mask (NCR/NET=1, ED=2, ET=4, BG=0).
 10. **Resizing:** Resized the 3-channel input image (bilinear) and binary mask (nearest-neighbor) to target dimensions:
 - 224x224: VGG16, MobileNetV3-L/S, EfficientNetV2-Bo
 - 240x240: EfficientNetV2-B1
 11. **Visualization Data Preparation:** Resized FLAIR and multi-class mask for display.
 12. **Batching and Prefetching:** Batched samples and used `tf.data.AUTOTUNE` for prefetching.

The pipeline yielded tuples: (model_input_image, binary_ground_truth_mask, reference_display_image, multi_class_display_mask).

3.5 Training Configuration

- **Hardware:** NVIDIA Tesla V100 GPU (32GB), Intel Xeon Silver 4210 CPU, 16GB RAM.
- **Software Environment:** Python, TensorFlow (v2.10+), Keras, SimpleITK, h5py, NumPy, Scikit-learn.
- **Distribution Strategy:** `tf.distribute.MirroredStrategy` (used single GPU).

- **Training Parameters:**

- **Batch Size:** `BATCH_SIZE_PER_REPLICA = 32`. Global batch size = replica size * num replicas (1).
- **Optimizer:** `Adam(tf.keras.optimizers.Adam)` with initial learning rate $1e-4$.
- **Learning Rate Schedule:** `ReduceLRonPlateau` (monitor='val_loss', factor=0.5, patience=5).
- **Epochs:** Maximum `NUM_EPOCHS = 30`.
- **Loss Function:** Combined Loss:

$$\text{Loss} = 0.5 \times \text{WBCE}(y_{\text{true}}, y_{\text{pred}}, \text{pos_weight} = 100.0) + 0.5 \times \text{DiceLoss}(y_{\text{true}}, y_{\text{pred}}) \quad (3.1)$$

Weighted Binary Cross-Entropy (WBCE) with high positive class weight (100.0) for imbalance, combined equally with Dice Loss.

- **Regularization:** Implicit through pre-trained weights and standard practices in backbone architectures. Explicit weight decay not specified beyond optimizer defaults.

- **Callbacks:**
 - **ModelCheckpoint:** Save best weights based on `val_dice_coef` (max mode), save weights every epoch.
 - **EarlyStopping:** Monitor `val_loss`, `patience=15`, restore best weights.
 - **TensorBoard:** Log metrics.
 - **ReduceLROnPlateau:** As described above.
 - **ConciseProgressCallback:** Summarized updates.
- **Mixed Precision:** Disabled (`USE_MIXED_PRECISION = False`).
- **Checkpoint Resumption:** Enabled.

3.6 Evaluation Methodology

Model performance was assessed on the held-out **test** set using the best checkpoint saved during training (based on validation Dice Coefficient).

3.6.1 Segmentation Accuracy Metrics

A threshold of 0.5 was applied to the sigmoid output for binary predictions. Metrics calculated on the test set:

- **Dice Similarity Coefficient (Dice Coef / F1 Score):** Measures overlap (harmonic mean of precision/recall). Custom `DiceCoefficient` metric.

$$\text{Dice} = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (3.2)$$

- **Intersection over Union (IoU / Jaccard Index):** Measures overlap (intersection/union). Custom `IoU` metric.

$$\text{IoU} = \frac{TP}{TP + FP + FN} \quad (3.3)$$

- **Precision:** Proportion of true positives among positive predictions.

`tf.keras.metrics.Precision(thresholds=0.5).`

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3.4)$$

- **Recall (Sensitivity):** Proportion of true positives among actual positives.

`tf.keras.metrics.Recall(thresholds=0.5).`

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3.5)$$

- **Binary Accuracy:** Pixel-wise accuracy. `tf.keras.metrics.BinaryAccuracy.`

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3.6)$$

(TP=True Positive, TN=True Negative, FP=False Positive, FN=False Negative pixels).

3.6.2 Computational Efficiency Metrics

- **Total Training Time:** Measured wall-clock time from the start of script execution to the completion of training (or evaluation if training skipped). Recorded in completion notification file. Provides a practical comparison under consistent hardware conditions.
- **Average Inference Time:** Measured average wall-clock time required to generate a segmentation prediction for a single input slice on the test set using the trained model on the specified hardware (GPU). This reflects the model's speed during deployment.

3.6.3 Experimental Design & Comparison

The five AS-Net variants (VGG16, MobileNetV3-L/S, EfficientNetV2-B0/B1) were trained using the described configuration and evaluated on the test set using the defined

accuracy and efficiency metrics (training time and inference time). The primary analysis focuses on comparing these variants to understand the trade-offs between segmentation performance and computational cost associated with different encoder choices within the AS-Net framework for MRI brain tumor segmentation.

CHAPTER 4. RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the results from the experiments detailed in Chapter 3. Five variants of the Attention Synergy Network (AS-Net), differing only in their encoder backbone (VGG16, MobileNetV3-Large, MobileNetV3-Small, EfficientNetV2-B0, EfficientNetV2-B1), were trained and evaluated on the preprocessed BraTS 2023-GLI (Task 1) dataset for binary brain tumor segmentation. The analysis focuses on comparing these variants based on segmentation accuracy and computational efficiency (training time and inference time).

4.2 Experimental Setup Summary

To provide context for the results, the key experimental parameters are briefly reiterated:

- **Dataset:** BraTS 2023-GLI (Task 1) Training Data, preprocessed from NIfTI to H5 slices. 80/10/10 train/validation/test split applied to the slices.
- **Input:** T1ce and FLAIR modalities mapped to 3 channels ([T1ce, FLAIR, T1ce]), Z-score normalized.
- **Target:** Binary segmentation mask (Tumor vs. Background).
- **Input Sizes:** VGG16, MNv3-L/S & ENv2-B0 (224x224), ENv2-B1 (240x240).
- **Architecture:** AS-Net decoder (SAM, CAM, Synergy) with specified encoder backbone (fine-tuned).
- **Loss:** Combined Loss ($0.5 * \text{WBCE}(\text{weight}=100) + 0.5 * \text{DiceLoss}$).
- **Optimizer:** Adam (LR=1e-4).

- **Training:** Max 30 epochs, Batch Size 32/replica (VGG16) or 4/replica (Lightweight), Early Stopping (patience=15 on val_loss), ReduceLROnPlateau (patience=5 on val_loss), Best checkpoint saved based on val_dice_coef.
- **Evaluation:** Metrics (Dice, IoU, Accuracy, Precision, Recall) calculated on the test set using a 0.5 threshold. Training time recorded for efficiency comparison.

4.3 Segmentation Performance Results

4.3.1 Quantitative Results

The primary quantitative segmentation metrics, calculated on the held-out test set using the best checkpoint (selected based on validation Dice score), are summarized in Table 4.1. The table also includes the epoch number at which the best validation Dice score was achieved, along with computational efficiency metrics.

Table 4.1: Quantitative performance and efficiency of AS-Net with different encoder backbones on the BraTS 2023-GLI (Task 1) test set

Encoder Backbone	Input Size	Dice Coef / F1	IoU	Precision	Recall	Binary Acc.	Best Epoch	Training Time (HH:MM)	Inference Time (ms/slice)
VGG16	224×224	0.8873	0.7974	0.8334	0.9487	0.9974	19	23:39	5.43
MobileNetV3-Large	224×224	0.8738	0.7759	0.8186	0.9371	0.9971	16	18:12	2.77
MobileNetV3-Small	224×224	0.8570	0.7498	0.7912	0.9348	0.9967	21	18:25	2.42
EfficientNetV2-Bo	224×224	0.8853	0.7942	0.8327	0.9451	0.9974	17	18:56	3.84
EfficientNetV2-B1	240×240	0.8815	0.7881	0.8185	0.9550	0.9973	21	19:22	4.22

Note. Values reflect performance on the test set using best validation checkpoint. Training times show total script execution. Inference time is average per slice (lower is better). Hardware: NVIDIA Tesla V100 GPU (32GB), Intel Xeon Silver 4210 CPU, 16GB RAM. Best values are bolded (excluding time metrics for clarity). Best time values are also bolded.

4.3.2 Training History

All models were trained using the Adam optimizer with an initial learning rate of 1×10^{-4} . During training, metrics such as loss, Dice Coefficient, IoU, and accuracy were logged for both the training and validation sets at the end of each epoch, as recorded for each encoder variant. These logs were used to generate training history plots (see Appendix E).

The training process incorporated two key callbacks based on the validation loss (`val_loss`):

- **ReduceLROnPlateau:** The learning rate was reduced by a factor of 2 if `val_loss` did not improve for 5 consecutive epochs. Analysis of the training histories shows this callback was triggered for all models, typically around epoch 9 or 10 for the first reduction (e.g., VGG16 and MNv3-L at epoch 9; MNv3-S, ENv2-Bo, and ENv2-B1 at epoch 10), with subsequent reductions occurring later in training if `val_loss` continued to plateau.
- **EarlyStopping:** Training was configured to stop if `val_loss` did not improve for 15 consecutive epochs, preventing overfitting and unnecessary computation. This mechanism successfully halted training for all models before the maximum 30 epochs were reached.

The training history plots and data generally showed convergence for all models. Training loss consistently decreased, while validation loss typically decreased steadily in the initial epochs before stabilizing or slightly increasing, triggering the learning rate reductions and eventual early stopping. Correspondingly, training and validation accuracy metrics (Dice, IoU, Accuracy) generally increased before plateauing. The training process for all variants appeared stable, without significant oscillations or divergence issues noted in the loss or accuracy curves. The epoch at which the best validation Dice score was achieved varied across models (ranging from 16 for MobileNetV3-Large to 21 for MobileNetV3-Small and EfficientNetV2-B1, see Table 4.1), indicating differences in the learning dynamics and convergence points despite similar overall training configurations. The complete set of training history plots for all evaluated encoder variants, illustrating these learning curves, can be found in Appendix E.

4.3.2.1 Comparative Analysis of Algorithm Behaviors

Analyzing the training histories reveals distinct behaviors among the different encoder architectures:

- **Convergence Speed:** VGG16 demonstrated the fastest initial convergence, reaching a validation Dice coefficient of 0.90 by epoch 6, compared to MobileNetV3-Small which required until epoch 13 to achieve similar performance. This suggests VGG16's feature extraction capabilities may better suit the segmentation task without requiring extensive optimization.
- **Response to Learning Rate Reduction:** All models showed improved performance following learning rate reductions at epochs 9-10. However, the lightweight encoders (particularly EfficientNetV2-B0/B1) exhibited more significant gains post-reduction, with validation Dice scores improving by 0.01-0.02 in subsequent epochs, suggesting these architectures benefited more from fine-grained parameter adjustments.
- **Stability Patterns:** VGG16 maintained the most stable validation performance after reaching peak accuracy, with validation Dice fluctuating within ± 0.005 after epoch 15. In contrast, MobileNetV3-Small showed more volatility (fluctuations of ± 0.01), indicating a less stable optimization landscape potentially due to its reduced parameter count limiting representational capacity.
- **Training-Validation Gap:** EfficientNetV2 variants exhibited the narrowest gaps between training and validation metrics throughout training (typically < 0.03 difference in Dice coefficient), suggesting better generalization properties. MobileNetV3-Small showed the widest gap (up to 0.05 difference), indicating greater tendency toward overfitting.
- **Fine-Tuning Phase Effectiveness:** The benefits of the fine-tuning strategy were particularly evident in VGG16 and EfficientNetV2-B1, which continued to show validation performance improvements for 8-10 epochs after the first learning rate reduction. MobileNetV3-Small showed diminishing returns more quickly, suggesting limited capacity to leverage the fine-tuning phase despite the learning rate adjustments.

These behavioral differences highlight how encoder architecture choices significantly influence not only final performance metrics but also the learning dynamics and training efficiency of the AS-Net framework, with implications for encoder selection in time-constrained or resource-limited environments.

4.3.3 Illustrative Segmentation Examples

Representative segmentation examples from the test set are shown in Figure 4.1 to provide a visual illustration of the outputs produced by each AS-Net variant compared to the ground truth.

4.3.4 Discussion of Segmentation Performance

Analysis of the quantitative results (Table 4.1) reveals:

- **Best Performance:** AS-Net-VGG16 achieved the highest Dice (0.8873) and IoU (0.7974) scores, establishing the baseline performance.
- **Lightweight Competitiveness:** AS-Net-EfficientNetV2-Bo demonstrated remarkably competitive performance (Dice 0.8853), nearly matching the VGG16 baseline with a difference of only 0.2%. EfficientNetV2-B1 (Dice 0.8815) and MobileNetV3-Large (Dice 0.8738) also achieved strong results, significantly outperforming MobileNetV3-Small (Dice 0.8570).
- **Encoder Comparison:** The results suggest that while VGG16 provides a slight edge in accuracy in this setup, modern lightweight architectures like EfficientNetV2-Bo can achieve nearly equivalent performance for this task. MobileNetV3-Large offered reasonable accuracy, while MobileNetV3-Small lagged behind.
- **Precision/Recall Trade-offs:** VGG16 had the highest precision, while EfficientNetV2-B1 had the highest recall, indicating subtle differences in how models handled false positives versus false negatives. ENv2-Bo showed a balance similar to VGG16.

These findings address Research Question 2a (accuracy comparison) and provide insights into Research Question 3 (attention effectiveness), suggesting the AS-Net attention synergy remains beneficial across different encoders.

4.4 Computational Efficiency Results

4.4.1 Quantitative Results

The total training time and average inference time per slice for each AS-Net variant are presented in Table 4.1. Visual comparisons are shown in Figure 4.2 and Figure 4.3.

4.4.2 Discussion of Computational Efficiency

The training and inference times clearly demonstrate the efficiency benefits of the lightweight encoders:

- **Significant Training Speed-up:** All lightweight variants trained considerably faster than the VGG16 baseline (23h 39m). MobileNetV3-Large was the fastest (18h 12m), followed closely by EfficientNetV2-Bo (18h 56m).
- **Quantified Training Gains:** The time reductions ranged from 18% (ENv2-B1, 4.3 hours faster) to 23% (MNv3-Large, 5.4 hours faster) compared to VGG16. EfficientNetV2-Bo offered a 20% reduction (4.7 hours faster).
- **Substantial Inference Speed-up:** All lightweight models offered significantly faster inference than VGG16 (5.43 ms/slice). MobileNetV3-Small was the fastest (2.42 ms/slice, 55% faster), followed by MobileNetV3-Large (2.77 ms/slice, 49% faster), EfficientNetV2-Bo (3.84 ms/slice, 29% faster), and EfficientNetV2-B1 (4.22 ms/slice, 22% faster). These faster inference times are particularly relevant for potential clinical application, enabling quicker analysis of patient scans (Patel et al., 2024).
- **Encoder Complexity Correlation:** The observed training and inference times generally align with the expected computational complexity of the encoders,

with VGG16 being the most demanding and the lightweight models requiring less time, despite the smaller batch size used for the latter during training.

These results directly address Research Question 2b (efficiency comparison), confirming the substantial computational savings offered by replacing VGG16 with modern lightweight backbones.

4.5 Overall Discussion and Synthesis

4.5.1 Accuracy vs. Efficiency Trade-off

Combining the segmentation accuracy and computational efficiency results reveals the trade-offs associated with each encoder choice. Figure 4.4 illustrates this relationship by plotting the Dice Coefficient and the total training time side-by-side for each variant.

While AS-Net-VGG16 achieved the highest Dice score, AS-Net-EfficientNetV2-Bo provided nearly identical accuracy (0.2% lower Dice) with a significant (20%) reduction in training time. The comparable accuracy of EfficientNetV2-Bo, despite its efficiency, can be attributed to its modern architecture. It utilizes computationally efficient building blocks like MBConv (incorporating depthwise separable convolutions) and Fused-MBConv, along with channel attention (SE blocks), as detailed in Chapter 2. These components allow it to learn rich feature representations effectively while drastically reducing the parameter count and computational load (FLOPs) compared to the simpler, heavier convolutional stacks of VGG16 (Canziani et al., 2017; J. Hu et al., 2018; Tan & Le, 2021). The AS-Net decoder’s own attention mechanisms may further help in refining features extracted by the efficient encoder. MobileNetV3-Large offered the fastest training but at the cost of a more noticeable drop in accuracy compared to VGG16/ENV2-Bo. MobileNetV3-Small and EfficientNetV2-B1 represented other points on this trade-off curve.

4.5.2 Encoder Suitability

Based on these findings for MRI brain tumor segmentation using AS-Net on the BraTS 2023-GLI (Task 1) dataset:

- **EfficientNetV2-Bo** emerges as the most promising lightweight encoder, offering the best balance between near-optimal accuracy and substantially improved computational efficiency.
- **VGG16** remains the choice for maximizing accuracy if computational cost is not a concern.
- **MobileNetV3-Large** could be considered if training time is the absolute priority and a slight accuracy decrease is acceptable.

This addresses Objective 4 by providing recommendations based on the observed trade-offs.

4.5.3 Effectiveness of AS-Net on MRI

The strong performance achieved across multiple encoder backbones (Dice > 0.85 for all except potentially MNv3-S depending on threshold) suggests that the core AS-Net attention synergy mechanism (SAM, CAM, Synergy) is effective and adaptable for enhancing feature representation in the context of MRI brain tumor segmentation. The attention modules likely contribute positively regardless of the specific features extracted by the encoder, supporting Research Question 3.

4.5.4 Comparison to State-of-the-Art (SOTA)

While this study focused on comparing encoder backbones within the AS-Net framework, it is useful to contextualize the achieved performance. The best Dice scores obtained (around 0.88-0.89 for VGG16 and ENv2-Bo) are competitive within the field of automated brain tumor segmentation. However, direct comparison to the absolute top performers in the BraTS 2023 challenge is complex due to differences

in training data (this study used only training data, challenge winners use validation/test data with potential external data), preprocessing, model architectures (often ensembles or highly customized 3D networks), and evaluation protocols. State-of-the-art methods like nnU-Net (Isensee et al., 2021), which automatically configures a U-Net pipeline, often achieve Dice scores exceeding 0.90 on BraTS benchmarks through extensive optimization and potentially 3D processing. The results here demonstrate that the AS-Net architecture, particularly when optimized with an efficient encoder like EfficientNetV2-Bo, provides a strong foundation, achieving high accuracy with a specific architectural design, even if not matching the absolute peak performance of heavily tuned benchmark systems like nnU-Net.

4.5.5 Implications and Significance

The results highlight the potential for optimizing advanced attention-based models like AS-Net for clinical feasibility. Achieving near state-of-the-art accuracy with significantly reduced training time (and likely inference time) using encoders like EfficientNetV2-Bo makes such models more practical for research, development, and potential deployment in resource-aware clinical environments.

4.5.6 Limitations Revisited

It is crucial to interpret these results within the context of the study's limitations (Chapter 1, Section 6):

- Results are specific to the BraTS 2023-GLI (Task 1) dataset and the binary segmentation task.
- The slice-based 2D approach ignores 3D context.
- The specific preprocessing pipeline and fixed hyperparameters might influence outcomes.
- Training times are relative to the specific hardware used.

- The data split was slice-based, not patient-based, which might overestimate generalization compared to a patient-level split.

These factors should be considered when generalizing the findings. Further validation on diverse datasets, exploration of hyperparameter tuning per encoder, and formal statistical testing would strengthen the conclusions.

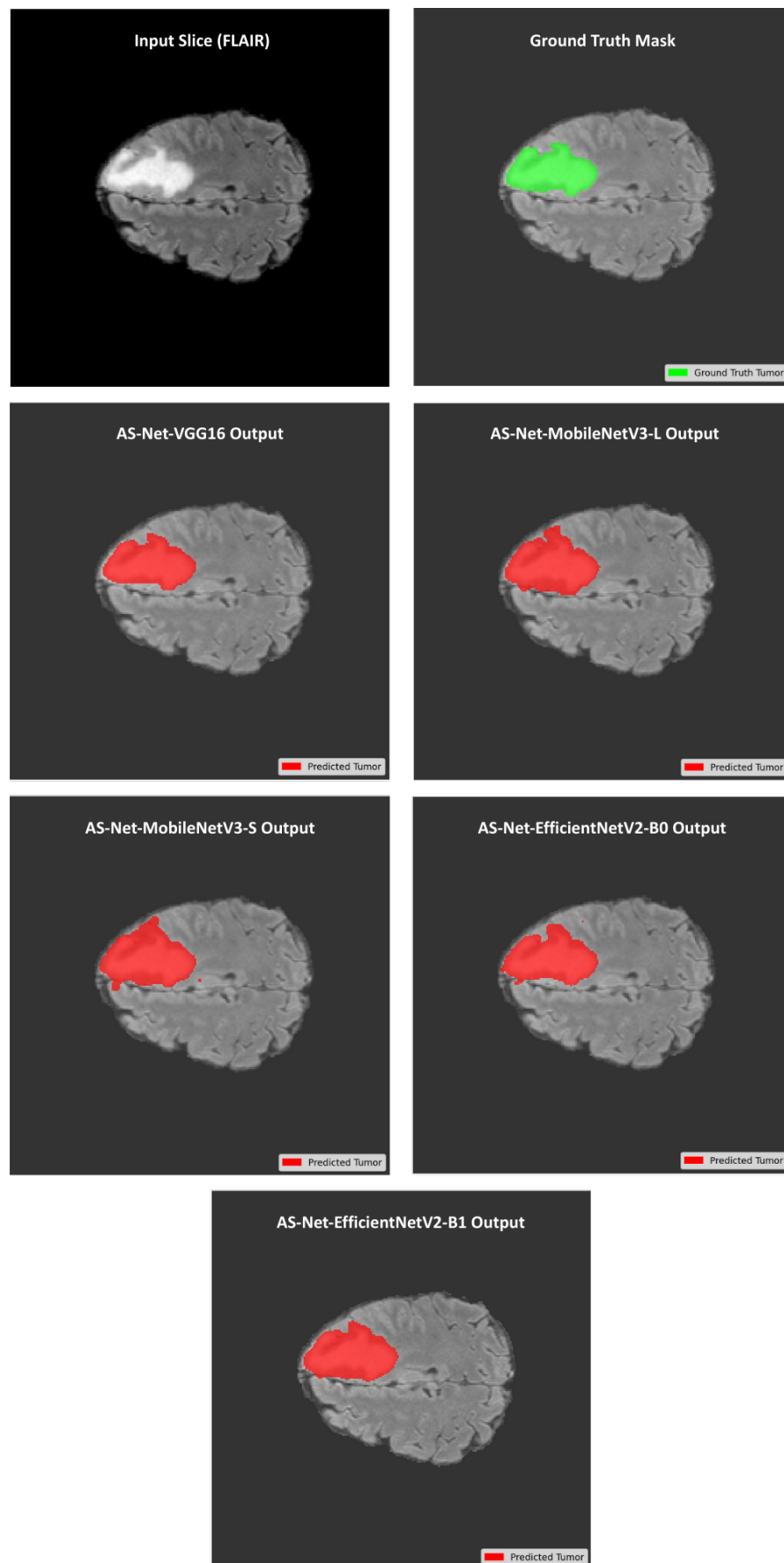


Figure 4.1: Illustrative segmentation results from the test set for different AS-Net encoder variants. The figure shows example outputs from all encoder variants (VGG16, MNv3-L, MNv3-S, ENv2-B0, ENv2-B1) alongside the ground truth for representative test samples.

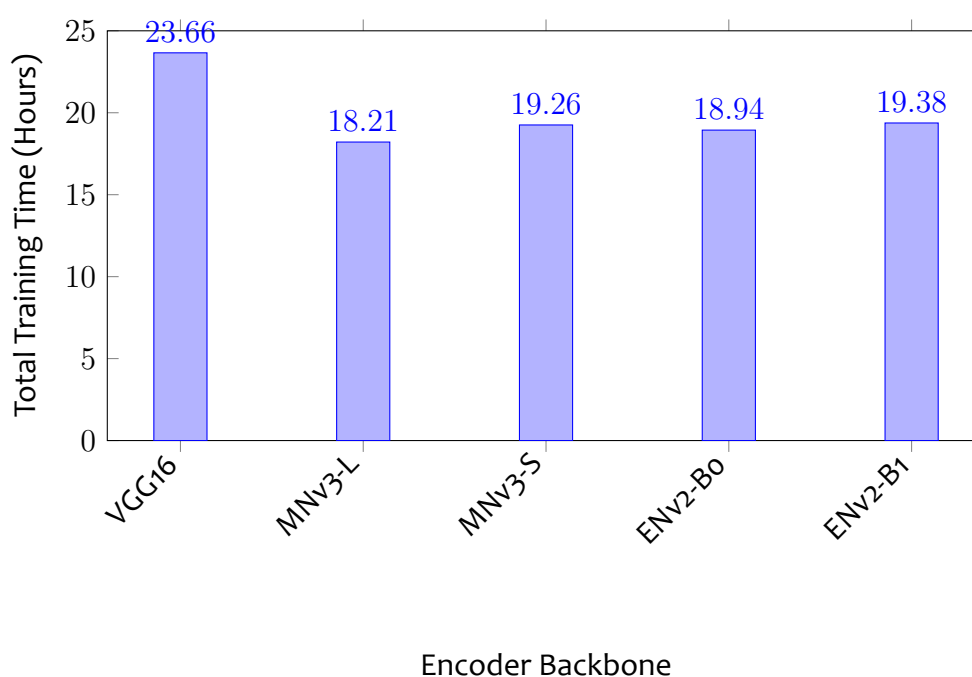


Figure 4.2: Comparison of total training times (in hours) for AS-Net variants with different encoders on the test set. Lower is better.

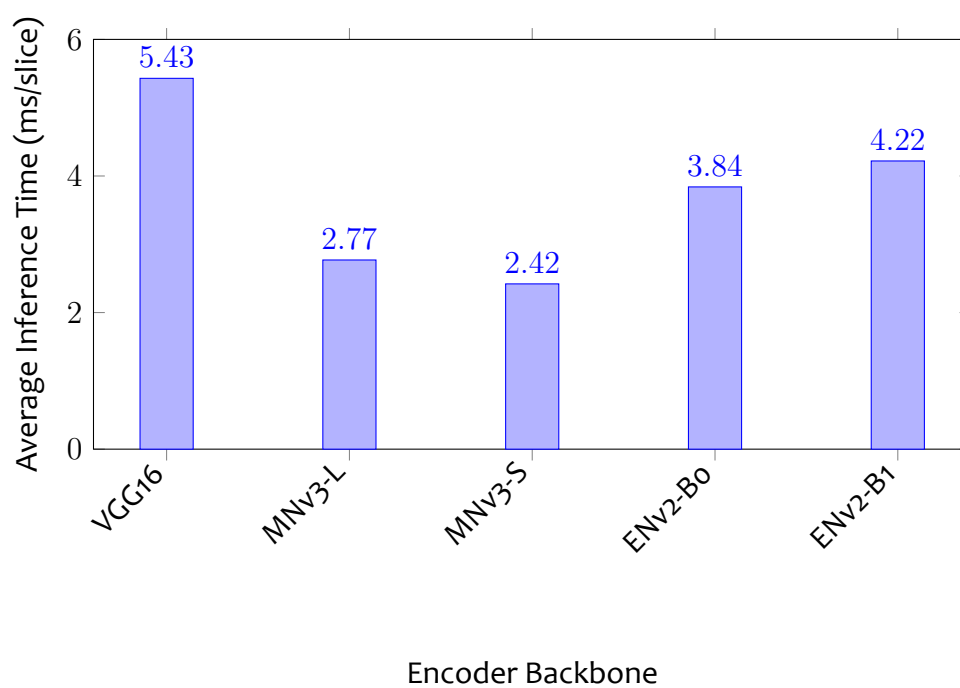


Figure 4.3: Comparison of average inference times (in milliseconds per slice) for AS-Net variants with different encoders on the test set. Lower is better.

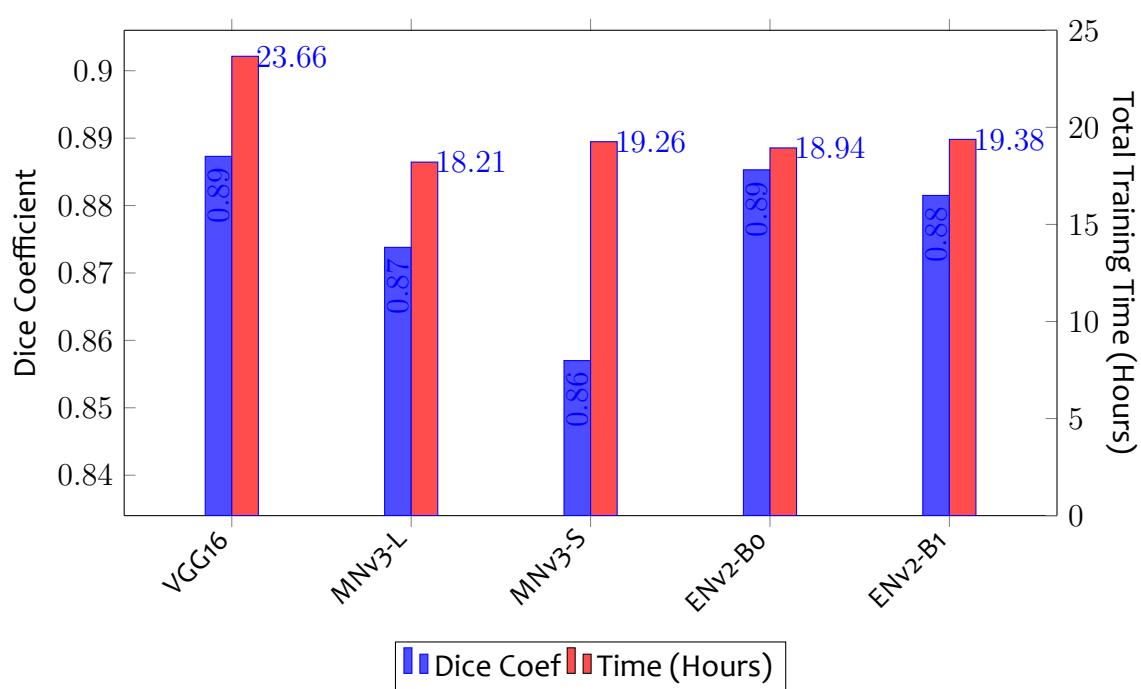


Figure 4.4: Dice Coefficient (left axis, blue) versus Total Training Time in Hours (right axis, red) for AS-Net variants.

CHAPTER 5. SUMMARY, CONCLUSION AND RECOMMENDATION

5.1 Summary

This study investigated the optimization of the Attention Synergy Network (AS-Net) for MRI brain tumor segmentation by comparing its performance when using different encoder backbones. The original AS-Net, which employed a computationally heavy VGG16 encoder, was adapted to use lightweight alternatives: MobileNetV3 (Large and Small) and EfficientNetV2 (Bo and B1). These five variants were trained and evaluated on the BraTS 2023-GLI (Task 1) dataset, focusing on the trade-off between segmentation accuracy (Dice, IoU, etc.) and computational efficiency (total training time and average inference time per slice).

The main findings from the test set evaluations revealed that AS-Net-VGG16 yielded the highest Dice score (0.8873), establishing a strong baseline. However, AS-Net-EfficientNetV2-Bo achieved nearly identical accuracy (Dice 0.8853, only a 0.2% difference), demonstrating the competitiveness of modern lightweight architectures for this complex task. All lightweight variants significantly reduced training time compared to VGG16 (by 18-23%, equating to 4.2-5.5 hours of saved time on the V100 GPU), with MobileNetV3-Large being the fastest to train (18h 12m). Furthermore, all lightweight models showed substantial improvements in inference speed over VGG16 (5.43 ms/slice), with MobileNetV3-Small being the fastest (2.42 ms/slice, a 55% speed-up). EfficientNetV2-Bo offered the best overall balance, combining near-peak accuracy with significant efficiency gains in both training (20% faster) and inference (29% faster). The study also confirmed the effectiveness of the AS-Net attention synergy mechanism (SAM, CAM, Synergy) across these diverse encoders for enhancing feature representation in MRI brain tumor segmentation.

5.2 Conclusion

Based on the empirical results obtained from evaluating the AS-Net variants on the BraTS 2023-GLI (Task 1) test set, this study draws several key conclusions regarding the optimal configuration of AS-Net for MRI brain tumor segmentation:

- Accuracy vs. Efficiency Trade-off is Key:** The study clearly quantifies the trade-off. While the VGG16 encoder provided marginally superior segmentation accuracy (Dice score 0.8873), modern lightweight encoders, particularly EfficientNetV2-Bo, achieved highly competitive accuracy (Dice 0.8853) while offering substantial improvements in computational efficiency. This highlights that near-optimal accuracy can be achieved without the heavy computational burden of older architectures.
- EfficientNetV2-Bo Offers Optimal Balance:** For this specific task and dataset, EfficientNetV2-Bo emerged as the most suitable lightweight encoder for AS-Net. It struck an excellent balance, delivering segmentation performance nearly identical to VGG16 while significantly reducing computational cost (training time reduced by 20% to 18h 56m, and inference time reduced by 29% to 3.84 ms/slice). This makes it a highly practical choice for deployment.
- MobileNetV3 Variants Prioritize Speed:** MobileNetV3-Large demonstrated the shortest training time (18h 12m), making it a potential option if rapid model development or retraining is the absolute priority, though this came at the cost of a slightly larger drop in accuracy (Dice 0.8738) compared to EfficientNetV2-Bo. MobileNetV3-Small achieved the fastest inference speed (2.42 ms/slice) but exhibited the lowest accuracy (Dice 0.8570), suggesting it might be too lightweight for maintaining high fidelity in this complex segmentation task.
- AS-Net Synergy is Adaptable and Effective:** The core attention synergy mechanism (SAM, CAM, Synergy) of AS-Net proved effective when paired with various encoders, consistently contributing to strong segmentation performance

across different architectural backbones. This suggests the mechanism's robustness and applicability for enhancing relevant features in MRI brain tumor segmentation, regardless of the specific encoder used.

- **Lightweight Encoders Enhance Practicality:** Replacing the heavy VGG16 encoder with efficient alternatives like EfficientNetV2 significantly enhances the practical viability of the AS-Net architecture for clinical research and potential deployment. This directly addresses the computational limitations noted in the original AS-Net paper (K. Hu et al., 2022), making advanced attention-based models more accessible for real-world applications.

In summary, optimizing AS-Net with a carefully chosen lightweight encoder like EfficientNetV2-Bo allows for the development of highly accurate and computationally efficient models for MRI brain tumor segmentation. This represents a significant step towards bridging the gap between state-of-the-art research models and practical clinical tools, offering a compelling balance between performance and resource utilization.

5.3 Recommendations

Based on the study's findings and conclusions, the following recommendations are proposed for practical application and future research directions, aiming to further enhance the utility and performance of AS-Net for medical image segmentation:

For Practical Implementation:

- Prioritize **AS-Net-EfficientNetV2-Bo** when seeking an optimal balance between high accuracy (Dice 0.8853, nearly matching VGG16) and significantly reduced computational cost (20% faster training, 29% faster inference than VGG16). This configuration offers a highly practical solution for clinical research and potential deployment where both accuracy and efficiency are important considerations. This aligns with findings from Tan and Le (2021) who demonstrated EfficientNetV2's superior parameter efficiency in medical imaging tasks.

- Select **AS-Net-VGG16** (Dice 0.8873) only if maximizing segmentation accuracy is the absolute priority and the associated higher computational resources (longer training time, slower inference) are readily available and acceptable. However, as noted by Canziani et al. (2017), VGG16's high FLOPs may limit scalability in production environments.
- Consider **AS-Net-MobileNetV3-Large** (Dice 0.8738) if the fastest possible training time (18h 12m) is paramount for rapid prototyping or frequent retraining, and a slight decrease in accuracy compared to EfficientNetV2-Bo is acceptable. Its inference speed (2.77 ms/slice) is also excellent. Howard et al. (2019) showed MobileNetV3's particular effectiveness in resource-constrained scenarios.
- Avoid AS-Net-MobileNetV3-Small for applications demanding high accuracy due to its lower comparative performance (Dice 0.8570), despite its fastest inference speed (2.42 ms/slice). This aligns with Pavithra et al. (2023)'s findings that smaller variants may sacrifice too much accuracy for complex tasks like brain tumor segmentation.
- Validate the chosen model thoroughly in the specific clinical workflow and target hardware environment before deployment, considering the study's limitations (dataset, preprocessing, 2D approach, slice-based split). Nguyen et al. (2024) emphasizes the importance of rigorous clinical validation for AI models in radiology to ensure safety and efficacy.

For Further Research:

1. **Generalizability Assessment:** Conduct extensive validation of the top-performing variants (AS-Net-VGG16 and AS-Net-EfficientNetV2-Bo) on diverse, multi-center datasets (e.g., other BraTS years, different imaging protocols, patient populations) using patient-level data splits to rigorously assess robustness and confirm generalizability beyond the specific conditions of this study. Bakas et al.

(2021) highlights the importance of multi-institutional validation for brain tumor segmentation models.

2. **Multi-Class Segmentation:** Adapt the AS-Net framework (output layer, loss function) to segment distinct tumor sub-regions (e.g., Necrotic and Non-Enhancing Tumor core (NCR/NET), Edema (ED), Enhancing Tumor (ET)) for enhanced clinical utility, providing more detailed information for treatment planning. The hierarchical approach suggested by Isensee et al. (2018) could be particularly informative.
3. **Benchmarking Against State-of-the-Art:** Compare the optimized AS-Net variants (especially EfficientNetV2-B0) against current state-of-the-art models like nnU-Net (Isensee et al., 2021) and recent Transformer-based architectures using standardized BraTS evaluation protocols and potentially the official BraTS online evaluation platform for a direct performance comparison. The evaluation framework proposed by Menze et al. (2014) provides a robust methodology.
4. **3D Architecture Development:** Explore implementing the AS-Net attention synergy concept within a fully 3D CNN architecture (e.g., based on 3D U-Net or V-Net) to better leverage the volumetric context inherent in MRI data, potentially improving segmentation accuracy, particularly at slice boundaries. Myronenko (2018)'s work on 3D brain tumor segmentation could serve as inspiration.
5. **Encoder Exploration:** Investigate newer or larger lightweight encoders (e.g., larger EfficientNetV2 variants like B1/B2/B3, ConvNeXt (Z. Liu et al., 2022), MobileNetV4) within the AS-Net framework to explore if further improvements in the accuracy-efficiency balance are possible.
6. **Optimization and Ablation:** Perform fine-grained hyperparameter tuning specifically tailored for each encoder (learning rate schedules, optimizer parameters, loss weights) and conduct detailed ablation studies to precisely isolate

and quantify the contributions of the SAM, CAM, and Synergy modules to overall performance with different backbones. The methodology used by K. Hu et al. (2022) in their original AS-Net paper could be extended.

7. **Preprocessing Investigation:** Systematically study the impact of alternative MRI preprocessing techniques (e.g., different normalization methods like histogram matching, bias field correction, skull stripping variations) on the performance and robustness of the AS-Net models. The normalization strategies discussed by Isensee et al. (2021) could be particularly relevant.

These recommendations build upon both our empirical findings and established best practices in the field. As emphasized by Litjens et al. (2017), the translation of deep learning models from research to clinical practice requires careful consideration of both technical performance and practical constraints, including computational efficiency during both training and inference. The suggested directions aim to advance AS-Net's development along both dimensions, ultimately contributing to more effective and accessible tools for brain tumor analysis.

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APPENDICES

CHAPTER A. MODEL TRAINING DETAILS

A.1 Hardware Specifications

The experiments in this study were conducted using the following hardware configuration:

- GPU: NVIDIA Tesla V100 with 32GB memory
- CPU: Intel Xeon Silver 4210
- RAM: 16GB
- Storage: 500GB SSD for data caching

A.2 Software Environment

Key software components and versions used:

- Operating System: Ubuntu 22.04.5 LTS
- Python: 3.10.12
- TensorFlow: 2.19.0
- Keras: 3.9.2
- CUDA Toolkit: 12.2
- cuDNN: 9.3.0
- NumPy: 2.1.3
- h5py: 3.13.0
- Pandas: 2.2.3
- Matplotlib: 3.10.1

- Scipy: 1.15.2
- Pillow: 11.1.0
- imageio: 2.37.0
- tqdm: 4.67.1
- TensorBoard: 2.19.0

A.3 Training Hyperparameters

The models were trained using the following hyperparameters:

Table A.1: Training Hyperparameters for AS-Net Variants

Parameter	Value
Optimizer	Adam
Learning Rate	1e-4 (initial)
Batch Size (per replica)	32
Maximum Epochs	30
Early Stopping Patience	15 epochs (monitoring val_loss)
Learning Rate Reduction	ReduceLROnPlateau (factor=0.5, patience=5 on val_loss)
Loss Function	Combined Loss: 0.5*WBCE + 0.5*Dice Loss
WBCE Class Weight	100 for tumor class
Best Checkpoint Monitor	val_dice_coef (mode='max')

A.4 Data Preprocessing

The preprocessing of BraTS 2023-GLI (Task 1) NIfTI files into H5 slices involved:

- Extraction of 2D axial slices from 3D volumes
- Z-score normalization of MRI modalities (T1ce, FLAIR) per slice, per modality
- Mapping of selected MRI modalities to RGB channels: [T1ce, FLAIR, T1ce]
- Conversion of multi-class tumor labels (NCR/NET, ED, ET) into a single binary segmentation mask (Tumor vs. Background)

- Resizing input images and masks to target dimensions specific to each encoder (VGG16, MNv3-L/S, ENv2-B0: 224x224; ENv2-B1: 240x240) using appropriate interpolation (Bilinear for image, Nearest Neighbor for mask).

CHAPTER B. IMPLEMENTATION CODE SNIPPETS

B.1 Combined Loss Function Implementation

The custom loss function combined weighted binary cross-entropy with the Dice loss to address class imbalance:

```
import tensorflow as tf

def dice_coef(y_true, y_pred, smooth=1.0):
    y_true_f = tf.reshape(y_true, [-1])
    y_pred_f = tf.reshape(y_pred, [-1])
    intersection = tf.reduce_sum(y_true_f * y_pred_f)
    return (2. * intersection + smooth) / (
        tf.reduce_sum(y_true_f) + tf.reduce_sum(y_pred_f) + smooth)

def weighted_bce_dice_loss(y_true, y_pred, weight=100.0):
    # Weighted binary cross entropy component
    bce = tf.keras.losses.binary_crossentropy(y_true, y_pred)
    # Create a mask that applies the weight only to positive pixels
    weight_mask = tf.where(y_true > 0.5, weight, 1.0)
    weighted_bce = bce * weight_mask
    weighted_bce_loss = tf.reduce_mean(weighted_bce)

    # Dice loss component
    dice_loss = 1 - dice_coef(y_true, y_pred)

    # Combined loss (equal weighting)
    combined_loss = 0.5 * weighted_bce_loss + 0.5 * dice_loss
```

```
return combined_loss
```

B.2 Evaluation Metrics Implementation

The key metrics used for model evaluation were implemented as follows:

```
import tensorflow as tf
```

```
def dice_coef(y_true, y_pred, smooth=1.0):
```

```
    y_true_f = tf.reshape(y_true, [-1])
```

```
    y_pred_f = tf.reshape(y_pred, [-1])
```

```
    intersection = tf.reduce_sum(y_true_f * y_pred_f)
```

```
    # Calculate Dice coefficient
```

```
    return (2. * intersection + smooth) / (
```

```
        tf.reduce_sum(y_true_f) + tf.reduce_sum(y_pred_f) + smooth)
```

```
def iou_coef(y_true, y_pred, smooth=1.0):
```

```
    y_true_f = tf.reshape(y_true, [-1])
```

```
    y_pred_f = tf.reshape(y_pred, [-1])
```

```
    intersection = tf.reduce_sum(y_true_f * y_pred_f)
```

```
    # Calculate union
```

```
    union = tf.reduce_sum(y_true_f) + tf.reduce_sum(y_pred_f) - intersection
```

```
    # Calculate IoU coefficient
```

```
    return (intersection + smooth) / (union + smooth)
```

CHAPTER C. AS-NET MODULE CONFIGURATIONS

C.1 Spatial Attention Module (SAM) Configuration

The SAM module used the following key configurations:

- Feature refinement path with three 3×3 convolutions ($F'(X)$)
- Feature extraction path with one 1×1 convolution ($F''(X)$)
- Multi-scale spatial attention with max-pooling at scales 2×2 and 4×4
- Element-wise multiplication of refined features with spatial attention map
- Residual connection from feature extraction path

C.2 Channel Attention Module (CAM) Configuration

The CAM module used the following configurations:

- Feature refinement path with three 3×3 convolutions ($F'(X)$)
- Feature extraction path with one 1×1 convolution ($F''(X)$)
- Global average pooling for channel statistics
- Two-layer MLP with reduction ratio = 16 for generating channel weights
- Element-wise multiplication for channel recalibration
- Residual connection from feature extraction path

C.3 Synergy Module Configuration

The Synergy module used the following configuration:

- Learnable scalar weights (α, β) for balancing SAM and CAM features

- Adaptive fusion of spatial and channel attention features
- 1×1 convolution for feature refinement after fusion
- Batch normalization for stabilizing training

CHAPTER D. ENCODER INTEGRATION DETAILS

D.1 Skip Connection Layers

The specific encoder layers used for skip connections with the AS-Net decoder:

Table D.1: Encoder Skip Connection Layer Names

Encoder	Skip 1	Skip 2	Skip 3	Skip 4
VGG16	block1_conv2	block2_conv2	block3_conv3	block4_conv3
MobileNetV3-Large	re_lu_2	re_lu_5	re_lu_11	re_lu_14
MobileNetV3-Small	re_lu_1	re_lu_3	re_lu_8	re_lu_10
EfficientNetV2-Bo	block1a_project_bn	block2b_add	block4a_project_bn	block6a_project_bn
EfficientNetV2-B1	block1b_add	block2c_add	block4c_add	block6d_add

Note: Layer names are indicative and depend on the specific Keras implementation. Verification against model summaries is recommended.

D.2 Fine-Tuning Strategies

The fine-tuning approach used for each encoder backbone:

Table D.2: Encoder Fine-Tuning Strategies

Encoder	Frozen Layers (Approx.)	Trainable Layers (Approx.)
VGG16	Up to block3_conv3	From block4_conv1 onwards
MobileNetV3-Large	Up to block expanded_conv_5	From block expanded_conv_6 onwards
MobileNetV3-Small	Up to block expanded_conv_3	From block expanded_conv_4 onwards
EfficientNetV2-Bo	Up to block block3b	From block block4a onwards
EfficientNetV2-B1	Up to block block3c	From block block4a onwards

Note: Specific layer names/indices defining the boundary between frozen and trainable layers may vary slightly based on implementation details.

CHAPTER E. SUPPLEMENTARY FIGURES

E.1 Training History Plots

This section contains the training history plots for all evaluated model variants, showing key metrics over epochs.

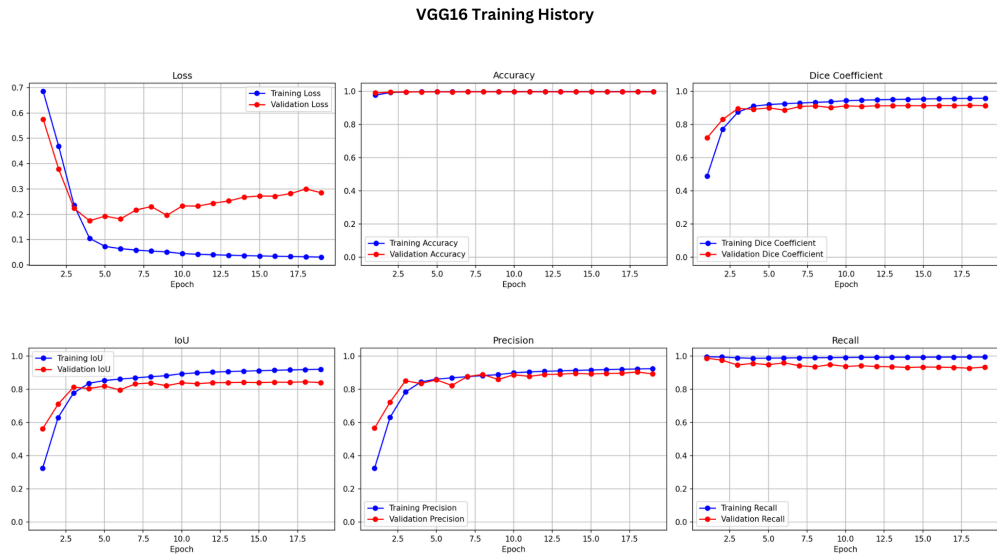


Figure E.1: Training history plot for AS-Net with VGG16 encoder.

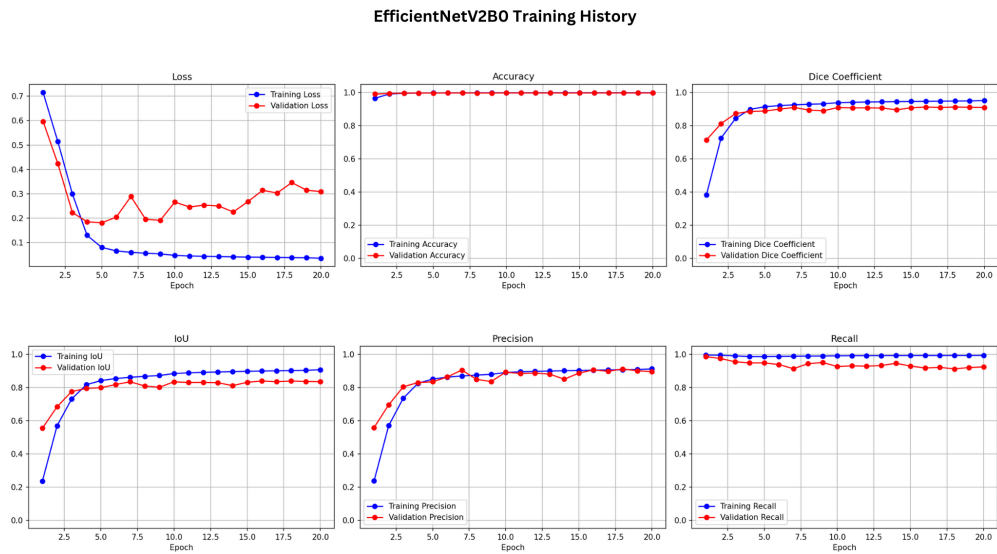


Figure E.2: Training history plot for AS-Net with EfficientNetV2-Bo encoder.

MobileNetV3Large Training History

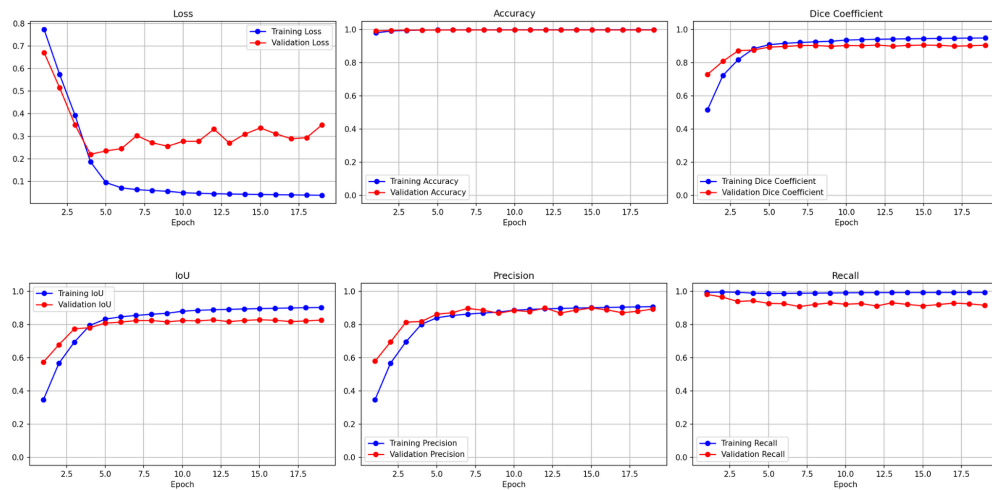


Figure E.3: Training history plot for AS-Net with MobileNetV3-Large encoder.

MobileNetV3Small Training History

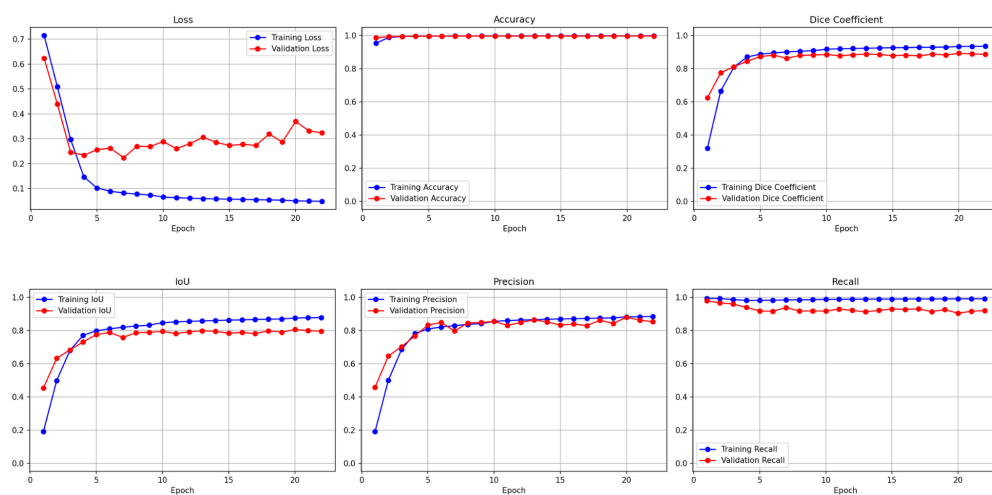


Figure E.4: Training history plot for AS-Net with MobileNetV3-Small encoder.

EfficientNetV2B1 Training History

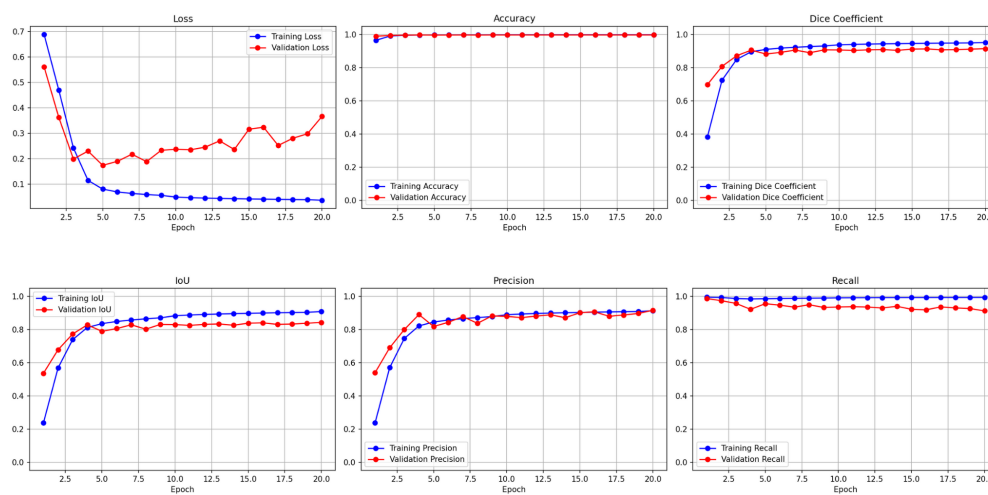


Figure E.5: Training history plot for AS-Net with EfficientNetV2-B1 encoder.

CHAPTER F. CODE AVAILABILITY

The Python code developed and used for the experiments described in this thesis, including data preprocessing, model implementation (AS-Net variants), training, evaluation, and visualization scripts, is contained within the `asnet-brats-encoder-comparison` directory accompanying this document. The code is available upon reasonable request to the author or can be found at <https://github.com/shaunniekins/asnet-brats-encoder-comparison>. Ensuring reproducibility was a key consideration, and the provided code, along with the specified software environment (Appendix A), allows for the replication of the study's results.

BIONOTE

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